

# Data Backup & Recovery Plan V3.3

Organization: Company A - Saudi Healthcare Authority  
Document ID: DOP-BCP-002  
Version: 3.3  
Classification: CONFIDENTIAL  
Date: September 15, 2024  
Total Pages: 32  
Status: APPROVED

## DOCUMENT CONTROL

| Version | Date               | Author            | Changes                                | Approver            |
|---------|--------------------|-------------------|----------------------------------------|---------------------|
| 1.0     | January 20, 2021   | Khalid Al-Mutairi | Initial backup plan creation           | Mohammed Al-Zahrani |
| 2.0     | August 15, 2022    | Khalid Al-Mutairi | Added cloud backup integration         | Mohammed Al-Zahrani |
| 2.5     | March 10, 2023     | Fahad Al-Dosari   | Ransomware immutability controls       | Omar Al-Rashid      |
| 3.0     | November 22, 2023  | Fahad Al-Dosari   | DR site procedures (Jeddah)            | Omar Al-Rashid      |
| 3.3     | September 15, 2024 | Fahad Al-Dosari   | Enhanced testing schedule and rollback | Omar Al-Rashid      |

Document Owner: IT Operations  
Review Cycle: Semi-Annual (Next Review: March 2025)  
Distribution: IT Operations, Database Administrators, Information Security, Business Continuity

## APPROVAL SIGNATURES

| Role                         | Name                 | Signature | Date               |
|------------------------------|----------------------|-----------|--------------------|
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| Information Security Officer | Omar Al-Rashid       | [Signed]  | September 14, 2024 |
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# 1. SCOPE & OBJECTIVES

## 1.1 Purpose

This plan establishes comprehensive procedures for backup and recovery operations to guarantee the recoverability of all business-critical systems in the event of:

- Hardware failure
- Software corruption
- Human error (accidental deletion)
- Cyberattack (Ransomware, malware)
- Natural disaster
- Facility failure (fire, flood, power)

## 1.2 Scope

This plan covers all systems within Company A's technology environment:

| System Category   | Systems Covered                      | Data Volume | Criticality |
|-------------------|--------------------------------------|-------------|-------------|
| Clinical Systems  | Epic EHR, Philips PACS, Sunquest LIS | 485 TB      | Tier 0-1    |
| Financial Systems | Oracle EBS, Payroll                  | 18 TB       | Tier 1      |
| HR Systems        | Oracle HCM                           | 1 TB        | Tier 2      |
| Infrastructure    | Active Directory, DNS, DHCP          | 500 GB      | Tier 0      |
| Collaboration     | SharePoint, Exchange Online          | 12 TB       | Tier 2      |
| Security          | Splunk SIEM, CyberArk PAM            | 8 TB        | Tier 1      |

## 1.3 Objectives

| Objective                    | Target                   | Current Status |
|------------------------------|--------------------------|----------------|
| Ensure data recoverability   | 100% of Tier 0-2 systems | Achieved       |
| Meet regulatory requirements | NDMO, CBAHI, ZATCA       | Compliant      |
| Minimize data loss           | Per RPO by tier          | Within targets |
| Minimize downtime            | Per RTO by tier          | Within targets |
| Protect against ransomware   | Immutable backups        | Implemented    |
| Enable geographic recovery   | DR site operational      | Operational    |

## 1.4 Regulatory Requirements

| Regulation     | Requirement                           | Implementation    |
|----------------|---------------------------------------|-------------------|
| NDMO Domain 4  | Data backup and recovery controls     | This plan         |
| NDMO Domain 10 | Business continuity for data services | DR site           |
| CBAHI          | Medical record availability           | Tier 1 RPO/RTO    |
| ZATCA          | Financial record preservation         | 10-year retention |
| PDPL           | Personal data protection              | Encrypted backups |

# 2. RPO & RTO DEFINITIONS

## 2.1 Key Definitions

| Term                                          | Definition                                    | Business Impact                            |
|-----------------------------------------------|-----------------------------------------------|--------------------------------------------|
| RPO (Recovery Point Objective)                | Maximum acceptable data loss measured in time | How much data can we afford to lose?       |
| RTO (Recovery Time Objective)                 | Maximum acceptable downtime                   | How long can we be down?                   |
| MTPD (Maximum Tolerable Period of Disruption) | Beyond this, business viability threatened    | When does extended outage become critical? |
| MBCO (Minimum Business Continuity Objective)  | Minimum service level during recovery         | What must function during recovery?        |

## 2.2 System Tier Definitions

### ***Tier 0 - Vital (Infrastructure)***

- Definition: Systems without which no other systems can function
- Examples: Active Directory, DNS, Core Network
- Business Impact: Complete organizational paralysis

### ***Tier 1 - Critical (Core Business)***

- Definition: Systems essential for core healthcare delivery
- Examples: Epic EHR, Oracle EBS, PACS
- Business Impact: Cannot provide patient care or process payments

### ***Tier 2 - Important (Business Support)***

- Definition: Systems that support business operations
- Examples: Email, SharePoint, CRM
- Business Impact: Reduced productivity, delayed communications

### ***Tier 3 - Archive (Historical)***

- Definition: Systems containing historical/reference data
- Examples: File archives, historical reports
- Business Impact: Limited operational impact

## **2.3 RPO/RTO Matrix**

| Tier   | Criticality | RPO        | RTO        | MTPD     | Systems                             |
|--------|-------------|------------|------------|----------|-------------------------------------|
| Tier 0 | Vital       | 0 Minutes  | 15 Minutes | 1 Hour   | Active Directory, DNS, DHCP         |
| Tier 1 | Critical    | 15 Minutes | 4 Hours    | 24 Hours | Epic EHR, Oracle EBS, PACS, Payroll |
| Tier 2 | Important   | 4 Hours    | 24 Hours   | 72 Hours | Email, SharePoint, CRM, HR          |
| Tier 3 | Archive     | 24 Hours   | 72 Hours   | 1 Week   | File Archives, Historical Data      |

## **2.4 Financial Impact Analysis**

| Downtime Duration | Tier 0 Impact  | Tier 1 Impact  | Tier 2 Impact |
|-------------------|----------------|----------------|---------------|
| 1 Hour            | 500,000 SAR    | 150,000 SAR    | 25,000 SAR    |
| 4 Hours           | 2,000,000 SAR  | 600,000 SAR    | 100,000 SAR   |
| 24 Hours          | 12,000,000 SAR | 3,600,000 SAR  | 600,000 SAR   |
| 72 Hours          | 36,000,000 SAR | 10,800,000 SAR | 1,800,000 SAR |

## 3. BACKUP STRATEGY (3-2-1 RULE)

### 3.1 Core Strategy

Company A follows the industry-standard 3-2-1 backup rule enhanced with immutability:

| Rule          | Definition                       | Implementation                           |
|---------------|----------------------------------|------------------------------------------|
| 3 Copies      | Three copies of all data         | Production + Local Backup + Cloud Backup |
| 2 Media Types | Two different storage media      | Disk (Veeam) + Cloud (AWS S3)            |
| 1 Off-site    | One copy geographically separate | Jeddah DR Site + AWS Region              |
| +1 Immutable  | One copy that cannot be altered  | AWS S3 Object Lock (30-day WORM)         |

### 3.2 Backup Types

#### 3.2.1 Full Backup

| Attribute      | Value                            |
|----------------|----------------------------------|
| Description    | Complete copy of all data        |
| Frequency      | Weekly (Friday 01:00 AM)         |
| Retention      | 4 weeks on-site, 12 months cloud |
| Size (typical) | 100% of source                   |
| Recovery Speed | Fastest (single restore)         |

#### 3.2.2 Incremental Backup

| Attribute      | Value                            |
|----------------|----------------------------------|
| Description    | Only changes since last backup   |
| Frequency      | Daily 01:00 AM                   |
| Retention      | 30 days                          |
| Size (typical) | 2-5% of source                   |
| Recovery Speed | Requires full + all incrementals |

#### 3.2.3 Differential Backup

| Attribute      | Value                            |
|----------------|----------------------------------|
| Description    | Changes since last full backup   |
| Frequency      | Not used (incremental preferred) |
| Retention      | N/A                              |
| Size (typical) | Grows daily                      |
| Recovery Speed | Full + single differential       |

### 3.2.4 Transaction Log Backup

| Attribute      | Value                            |
|----------------|----------------------------------|
| Description    | Database transaction logs        |
| Frequency      | Every 15 minutes (Tier 0-1 only) |
| Retention      | 7 days                           |
| Size (typical) | Variable (activity dependent)    |
| Recovery Speed | Point-in-time recovery           |

## 3.3 Backup Infrastructure

### 3.3.1 Primary Backup Environment (Riyadh)

| Component            | Specification                | Purpose                  |
|----------------------|------------------------------|--------------------------|
| Veeam Backup Server  | 2x Dell R750 (Clustered)     | Backup orchestration     |
| Veeam Proxy Servers  | 4x Dell R650                 | Data movement            |
| Primary Repository   | NetApp AFF A400 (200TB)      | On-site backup storage   |
| Secondary Repository | NetApp FAS2720 (500TB)       | Extended retention       |
| Tape Library         | IBM TS4500 (2PB capacity)    | Long-term archive        |
| Network              | Dedicated 40Gbps backup VLAN | Backup traffic isolation |

### 3.3.2 Cloud Backup Environment (AWS)

| Component      | Specification              | Purpose                  |
|----------------|----------------------------|--------------------------|
| S3 Standard    | Riyadh Region (me-south-1) | Recent backups (30 days) |
| S3 Glacier     | Riyadh Region (me-south-1) | Long-term retention      |
| S3 Object Lock | WORM compliance mode       | Ransomware protection    |

|                |                       |                         |
|----------------|-----------------------|-------------------------|
| VPN Connection | 10Gbps Direct Connect | Secure data transfer    |
| Encryption     | AES-256 (AWS KMS)     | Data at rest protection |

3.3.3 DR Site Environment (Jeddah)

| Component            | Specification             | Purpose               |
|----------------------|---------------------------|-----------------------|
| Veeam Replica Server | Dell R750                 | VM replication target |
| Storage Array        | NetApp AFF A250 (100TB)   | Replica storage       |
| Compute              | VMware vSphere 8 Cluster  | Recovery compute      |
| Network              | 10Gbps WAN link to Riyadh | Replication traffic   |

3.4 Encryption Standards

| Data State      | Encryption Method | Key Management       |
|-----------------|-------------------|----------------------|
| In Transit      | TLS 1.3           | Certificate-based    |
| At Rest (Local) | AES-256           | Veeam Enterprise Key |
| At Rest (Cloud) | AES-256           | AWS KMS (CMK)        |
| At Rest (Tape)  | AES-256           | Hardware encryption  |

4. BACKUP SCHEDULES BY SYSTEM

4.1 Tier 0 - Vital Systems

4.1.1 Active Directory Domain Controllers

| Backup Type    | Schedule                | Retention | Target           |
|----------------|-------------------------|-----------|------------------|
| System State   | Daily 02:00             | 30 days   | Veeam Repository |
| Full VM Backup | Weekly (Saturday 01:00) | 12 weeks  | Veeam + S3       |
| Incremental    | Daily 01:00             | 30 days   | Veeam Repository |
| Replication    | Continuous (15 min RPO) | N/A       | Jeddah DR        |

4.1.2 DNS/DHCP Servers

| Backup Type          | Schedule                | Retention | Target           |
|----------------------|-------------------------|-----------|------------------|
| Full VM Backup       | Weekly (Saturday 02:00) | 8 weeks   | Veeam Repository |
| Incremental          | Daily 02:00             | 14 days   | Veeam Repository |
| Configuration Export | Daily 03:00             | 30 days   | SharePoint       |

## 4.2 Tier 1 - Critical Systems

### 4.2.1 Epic EHR (Oracle RAC)

| Backup Type              | Schedule              | Retention                      | Target           |
|--------------------------|-----------------------|--------------------------------|------------------|
| RMAN Full                | Weekly (Friday 01:00) | 4 weeks local, 12 months cloud | Veeam + S3       |
| RMAN Incremental Level 1 | Daily 01:00           | 30 days                        | Veeam Repository |
| Archive Log              | Every 15 minutes      | 7 days                         | Veeam Repository |
| DataGuard Standby        | Continuous sync       | N/A                            | Jeddah DR        |

Recovery Capability: Point-in-time recovery to any transaction within 7 days

### 4.2.2 Philips PACS (Imaging Archive)

| Backup Type        | Schedule                | Retention     | Target           |
|--------------------|-------------------------|---------------|------------------|
| Full Image Archive | Weekly (Saturday 22:00) | 4 weeks       | Veeam Repository |
| Incremental        | Daily 22:00             | 30 days       | Veeam Repository |
| Cloud Archive      | Monthly                 | 10 years      | S3 Glacier       |
| Replication        | Daily batch             | 24-hour delay | Jeddah DR        |

Special Consideration: Large dataset (445TB) - backup window extended to 8 hours

### 4.2.3 Oracle EBS (Financials)

| Backup Type        | Schedule              | Retention                     | Target           |
|--------------------|-----------------------|-------------------------------|------------------|
| RMAN Full          | Weekly (Friday 02:00) | 4 weeks local, 10 years cloud | Veeam + S3       |
| RMAN Incremental   | Daily 02:00           | 30 days                       | Veeam Repository |
| Archive Log        | Every 15 minutes      | 7 days                        | Veeam Repository |
| Application Export | Monthly (1st Sunday)  | 12 months                     | S3 Standard      |



#### 4.2.4 Splunk SIEM

| Backup Type   | Schedule              | Retention                 | Target           |
|---------------|-----------------------|---------------------------|------------------|
| Index Backup  | Daily 04:00           | 90 days hot, 7 years cold | S3 Glacier       |
| Configuration | Daily 03:00           | 30 days                   | Veeam Repository |
| Search Heads  | Weekly (Sunday 03:00) | 8 weeks                   | Veeam Repository |

### 4.3 Tier 2 - Important Systems

#### 4.3.1 Microsoft 365 (Exchange Online, SharePoint)

| Backup Type       | Schedule    | Retention | Target         |
|-------------------|-------------|-----------|----------------|
| Mailbox Backup    | Daily 05:00 | 90 days   | Veeam for M365 |
| SharePoint Backup | Daily 06:00 | 90 days   | Veeam for M365 |
| OneDrive Backup   | Daily 07:00 | 90 days   | Veeam for M365 |
| Teams Backup      | Daily 08:00 | 90 days   | Veeam for M365 |

Note: Microsoft native retention + Veeam provides comprehensive protection

#### 4.3.2 Oracle HCM (HR System)

| Backup Type   | Schedule                | Retention | Target           |
|---------------|-------------------------|-----------|------------------|
| Full Database | Weekly (Saturday 03:00) | 4 weeks   | Veeam Repository |
| Incremental   | Daily 03:00             | 14 days   | Veeam Repository |
| Cloud Backup  | Weekly                  | 7 years   | S3 Standard      |

### 4.4 Backup Schedule Summary

| Day       | 01:00             | 02:00           | 03:00         | 04:00        | 05:00-08:00 |
|-----------|-------------------|-----------------|---------------|--------------|-------------|
| Sunday    | Incremental (All) | AD System State | Splunk Config | Splunk Index | M365 Backup |
| Monday    | Incremental (All) | AD System State | Splunk Config | Splunk Index | M365 Backup |
| Tuesday   | Incremental (All) | AD System State | Splunk Config | Splunk Index | M365 Backup |
| Wednesday | Incremental (All) | AD System State | Splunk Config | Splunk Index | M365 Backup |
| Thursday  | Incremental (All) | AD System State | Splunk Config | Splunk Index | M365 Backup |



- Notify Clinical Systems team

↓

Step 2: Connect to RMAN

```
$ rman target / catalog rman_user/password@rman_catalog
```

↓

Step 3: Identify available backups

```
RMAN> LIST BACKUP SUMMARY;
```

```
RMAN> LIST BACKUP OF DATABASE;
```

↓

Step 4: Restore database to point-in-time

```
RMAN> RUN {
```

```
SET UNTIL TIME "TO_DATE('2024-09-15 08:00:00','YYYY-MM-DD HH24:MI:SS)";
```

```
RESTORE DATABASE;
```

```
RECOVER DATABASE;
```

```
}
```

↓

Step 5: Open database with RESETLOGS

```
RMAN> ALTER DATABASE OPEN RESETLOGS;
```

↓

Step 6: Verify database integrity

```
$ sqlplus / as sysdba
```

```
SQL> SELECT COUNT() FROM patients;
```

```
SQL> SELECT MAX(encounter_date) FROM encounters;
```

↓

Step 7: Notify stakeholders

- Send recovery completion notification

- Document recovery in incident ticket

### 5.2.3 Estimated Recovery Time

| Scenario                  | Data Volume | Estimated Time |
|---------------------------|-------------|----------------|
| Single tablespace         | 50 GB       | 30 minutes     |
| Partial database          | 500 GB      | 2 hours        |
| Full database             | 35 TB       | 8-12 hours     |
| Point-in-time (with logs) | Variable    | +1 hour        |

## 5.3 Runbook: VM Recovery (Veeam)

### 5.3.1 Instant VM Recovery (Fastest)

Step 1: Open Veeam Backup & Replication Console

↓

Step 2: Navigate to Home > Backups > Disk

↓

Step 3: Right-click target VM > Instant Recovery > Instant Recovery to VMware

↓

Step 4: Select restore point (date/time)

↓

Step 5: Choose destination:

- Original location (replace existing)
- New location (for testing)

↓

Step 6: Configure network:

- Connect to original network (production)
- Connect to isolated network (testing)

↓

Step 7: Click "Finish" to start Instant VM Recovery

↓

Step 8: VM boots from backup storage (NFS mount)

- Immediate availability
- Performance may be reduced

↓

Step 9: Migrate to production storage

- Right-click running VM > "Migration to production"
- Storage vMotion to production datastore

Instant VM Recovery SLA: VM available within 5 minutes

### **5.3.2 Full VM Restore**

Step 1: Open Veeam Console > Home > Backups > Disk

↓

Step 2: Right-click target VM > Restore entire VM

↓

Step 3: Select restore point

↓

Step 4: Choose restore mode:

- Restore to original location
- Restore to new location

↓

Step 5: Configure options:

- Power on VM after restore: Yes/No
- Restore VM tags: Yes

↓

Step 6: Review and start restore

↓

Step 7: Monitor progress in Veeam console

↓

Step 8: Verify VM functionality post-restore

## 5.4 Runbook: File-Level Recovery

### 5.4.1 Windows File Recovery

Step 1: Open Veeam Console > Home > Backups

↓

Step 2: Right-click target VM > Restore guest files > Microsoft Windows

↓

Step 3: Select restore point

↓

Step 4: Browse backup contents (virtual file system mount)

↓

Step 5: Navigate to required files/folders

↓

Step 6: Right-click > Restore:

- Restore to original location
- Copy to different location
- Keep both (restore with suffix)

↓

Step 7: Verify file integrity

### 5.4.2 Oracle Data Pump Recovery (Table Level)

Step 1: Identify required backup

```
$ ls -la /backup/oracle/exports/
```

↓

Step 2: Import specific table

```
$ impdp system/password DIRECTORY=backup_dir \  
DUMPFILE=export_20240915.dmp \  
TABLES=schema.table_name \  
REMAP_TABLE=schema.table_name:schema.table_name_restored
```

↓

Step 3: Verify imported data

```
SQL> SELECT COUNT() FROM schema.table_name_restored;
```

↓

Step 4: Swap tables if needed

```
SQL> RENAME table_name TO table_name_old;  
SQL> RENAME table_name_restored TO table_name;
```

## 5.5 Runbook: Microsoft 365 Recovery

### 5.5.1 Mailbox Item Recovery

Step 1: Open Veeam Backup for Microsoft 365 Console

↓

Step 2: Navigate to Organizations > Company A

↓

Step 3: Right-click backup > Restore > Exchange items

↓

Step 4: Search for items:

- By user mailbox
- By date range
- By subject/content

↓

Step 5: Select items to restore

↓

Step 6: Choose restore target:

- Restore to original location
- Restore to different mailbox
- Export to PST

↓

Step 7: Execute and verify

## 6. RANSOMWARE PROTECTION

### 6.1 Immutable Backup Strategy

| Layer     | Protection Method         | Implementation           |
|-----------|---------------------------|--------------------------|
| Primary   | Air-gapped tape           | IBM TS4500 (offline)     |
| Secondary | S3 Object Lock            | AWS WORM compliance mode |
| Tertiary  | Veeam Hardened Repository | Linux immutable flag     |

### 6.2 AWS S3 Object Lock Configuration

```
json
{
  "Rule": {
```

```
"ObjectLockEnabled": "Enabled",
"DefaultRetention": {
"Mode": "COMPLIANCE",
"Days": 30
}
}
}
```

Effect: No one (including root/admin) can delete or modify backups for 30 days

## 6.3 Veeam Hardened Repository

| Setting     | Value                          |
|-------------|--------------------------------|
| OS          | Ubuntu 22.04 LTS (hardened)    |
| File System | XFS with immutable flag        |
| Network     | Isolated backup VLAN           |
| Access      | Single-use credentials per job |
| Monitoring  | CrowdStrike Falcon EDR         |

## 6.4 Ransomware Recovery Procedure

Step 1: ISOLATE - Disconnect affected systems from network

↓

Step 2: ASSESS - Determine scope of encryption

- Which systems affected?
- What is the encryption date/time?

↓

Step 3: IDENTIFY - Find last clean backup

- Check backup integrity (Veeam SureBackup)
- Verify no ransomware in backup

↓

Step 4: CLEAN - Rebuild infected systems

- Wipe and reinstall OS
- Restore from verified clean backup

↓

Step 5: RESTORE - Sequential recovery by tier

- Tier 0 first (AD, DNS)
- Then Tier 1 (EHR, EBS)
- Finally Tier 2

↓

Step 6: VERIFY - Confirm systems operational

- Application testing
- Data integrity checks



Step 7: RECONNECT - Gradually restore network

- Monitor for re-infection

## 7. DISASTER RECOVERY SITE

### 7.1 Jeddah DR Site Overview

| Attribute             | Details                     |
|-----------------------|-----------------------------|
| Location              | Jeddah Data Center (JDC-01) |
| Distance from Primary | 950 km                      |
| Connectivity          | 10 Gbps dedicated WAN       |
| Activation Time       | 4 hours for Tier 1 systems  |

### 7.2 Replicated Systems

| System           | Replication Method  | RPO      | RTO      |
|------------------|---------------------|----------|----------|
| Active Directory | DFS Replication     | 15 min   | 15 min   |
| Epic EHR         | Oracle DataGuard    | 15 min   | 4 hours  |
| Oracle EBS       | Oracle DataGuard    | 15 min   | 4 hours  |
| Philips PACS     | Veeam Replication   | 24 hours | 8 hours  |
| SharePoint       | Azure Site Recovery | 4 hours  | 24 hours |

### 7.3 DR Activation Procedure

Step 1: DR Declaration by authorized personnel:

- CEO, CIO, or IT Director



Step 2: Notify DR team and assemble



Step 3: Activate network failover:

- DNS updates (reduced TTL pre-configured)
- VPN gateway activation



↓

Step 4: Bring up Tier 0 systems:

- Promote DR Domain Controllers
- Activate DNS/DHCP

↓

Step 5: Activate Tier 1 systems:

- Failover Oracle DataGuard (Epic, EBS)
- Start application servers

↓

Step 6: Verify connectivity:

- Test critical workflows
- Confirm data integrity

↓

Step 7: Communicate status to organization

## 7.4 DR Site Capacity

| Resource       | Primary Site | DR Site | DR Capacity |
|----------------|--------------|---------|-------------|
| Compute (vCPU) | 2,400        | 1,200   | 50%         |
| Memory (TB)    | 12           | 6       | 50%         |
| Storage (TB)   | 600          | 200     | 33%         |
| Network (Gbps) | 100          | 40      | 40%         |

Note: DR site sized for critical systems only; non-critical systems suspended during DR

## 8. TESTING SCHEDULE & PROCEDURES

### 8.1 Testing Calendar

| Test Type           | Frequency     | Next Scheduled    | Responsible   |
|---------------------|---------------|-------------------|---------------|
| File Restore Test   | Monthly       | October 15, 2024  | Backup Admin  |
| Application Restore | Quarterly     | November 1, 2024  | DBA Team      |
| Full DR Simulation  | Annually      | February 15, 2025 | IT Operations |
| Ransomware Recovery | Semi-annually | December 1, 2024  | Security + IT |

### 8.2 Monthly File Restore Test

### 8.2.1 Test Procedure

Step 1: Select 5 random files from different systems

↓

Step 2: Request restore via ServiceNow

↓

Step 3: Execute restore to test location

↓

Step 4: Verify file integrity:

- Checksum comparison
- File opens correctly

↓

Step 5: Document results in test log

↓

Step 6: Report any failures for remediation

### 8.2.2 Success Criteria

| Metric               | Target       | Current   |
|----------------------|--------------|-----------|
| Restore Success Rate | 100%         | 100%      |
| Average Restore Time | < 15 minutes | 8 minutes |
| Data Integrity       | 100% match   | 100%      |

## 8.3 Quarterly Application Restore Test

### 8.3.1 Test Scope

| Quarter | System       | Test Type                                 |
|---------|--------------|-------------------------------------------|
| Q1      | Epic EHR     | Full database restore to test environment |
| Q2      | Oracle EBS   | Point-in-time recovery                    |
| Q3      | Philips PACS | Image archive restore                     |
| Q4      | All Tier 1   | Integrated recovery test                  |

### 8.3.2 Q4 Test Procedure (Integrated)

Step 1: Notify all stakeholders (2 weeks advance)

↓

Step 2: Schedule maintenance window (Saturday 06:00-18:00)

↓

Step 3: Execute sequential restore:

- Active Directory (isolated test domain)
- Epic EHR database
- Oracle EBS database
- Dependent applications

↓

Step 4: Validate application functionality:

- Login tests
- Transaction processing
- Report generation

↓

Step 5: Measure and document:

- Actual RTO achieved
- Data integrity verification
- Issues encountered

↓

Step 6: Cleanup test environment

## 8.4 Annual DR Simulation

### 8.4.1 Simulation Scope

- Full activation of Jeddah DR site
- All Tier 0 and Tier 1 systems
- 4-hour operational period
- Live transaction processing (limited)

### 8.4.2 Success Criteria

| Metric              | Target     | Last Test Result |
|---------------------|------------|------------------|
| Tier 0 RTO          | 15 minutes | 12 minutes       |
| Tier 1 RTO          | 4 hours    | 3.5 hours        |
| Data Loss (RPO)     | 15 minutes | 8 minutes        |
| Transaction Success | > 95%      | 98%              |

## 9. ROLES AND RESPONSIBILITIES

### 9.1 RACI Matrix

| Activity               | Backup Admin | DBA | IT Security | IT Director |
|------------------------|--------------|-----|-------------|-------------|
| Daily backup execution | R            | I   | I           | I           |
| Backup monitoring      | R/A          | I   | C           | I           |
| Recovery execution     | R            | R   | C           | A           |
| DR activation          | R            | R   | R           | A           |
| Testing coordination   | R            | C   | C           | A           |
| Policy approval        | I            | I   | C           | A           |

R = Responsible, A = Accountable, C = Consulted, I = Informed

## 9.2 On-Call Schedule

| Role          | Primary           | Secondary        | Escalation  |
|---------------|-------------------|------------------|-------------|
| Backup Admin  | Khalid Al-Mutairi | Fahad Al-Dosari  | IT Director |
| Oracle DBA    | Tariq Al-Rasheed  | Yusuf Al-Qahtani | DBA Manager |
| Windows Admin | Ahmad Al-Shehri   | Hassan Al-Ghamdi | IT Director |
| Network       | Faisal Al-Harbi   | Saeed Al-Malki   | IT Director |

## APPENDIX A: CONTACT ESCALATION MATRIX

| Level     | Time           | Contact               | Phone            |
|-----------|----------------|-----------------------|------------------|
| L1        | 0-15 min       | On-call Backup Admin  | +966-5X-XXX-XX01 |
| L2        | 15-30 min      | On-call DBA           | +966-5X-XXX-XX02 |
| L3        | 30-60 min      | IT Operations Manager | +966-5X-XXX-XX03 |
| L4        | 60+ min        | IT Director           | +966-5X-XXX-XX04 |
| Executive | Major incident | CIO                   | +966-5X-XXX-XX05 |

## APPENDIX B: BACKUP VERIFICATION CHECKLIST

### Daily Checklist

| Check                       | Status | Notes |
|-----------------------------|--------|-------|
| All backup jobs completed   | ■      |       |
| No errors in Veeam console  | ■      |       |
| Storage capacity > 20% free | ■      |       |
| Replication jobs successful | ■      |       |
| Cloud sync completed        | ■      |       |

## Weekly Checklist

| Check                          | Status | Notes |
|--------------------------------|--------|-------|
| Full backups completed         | ■      |       |
| Tape library status normal     | ■      |       |
| DR site connectivity verified  | ■      |       |
| SureBackup verification passed | ■      |       |
| Backup reports reviewed        | ■      |       |

Document End

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Version 3.3 | September 2024